Lecture 3: Metric Spaces and Topology

Mathematical Analysis

1 Metric Spaces

The four *metric axioms* are the following requirements on a function $d : X \times X \to \mathbb{R}$, where X is a set

Axiom 1.1 (Nonnegativity). [axm:metric-nonnegativity]

$$\forall x, y \in X, \quad d(x, y) \geq 0$$

Axiom 1.2 (Identity of Indiscernibles). [axm:metric-separation]

$$\forall x, y \in X, \quad d(x, y) = 0 \iff x = y$$

Axiom 1.3 (Symmetry). [axm:metric-symmetry]

$$\forall x, y \in X, \quad d(x, y) = d(y, x)$$

Axiom 1.4 (Triangle Inequality). [axm:metric-triangle-inequality]

$$\forall x, y, z \in X, \quad d(x, z) \leq d(x, y) + d(y, z)$$

Definition 1.1 (Metric). [def:metric] Let X be a set. A *metric*, or *distance function*, on X is a function $d : X \times X \to \mathbb{R}$ satisfying nonnegativity, identity of indiscernibles, symmetry, and the triangle inequality

Definition 1.2 (Metric Space). [def:metric-space] A *metric space* is a pair (X, d), where X is a set and d is a metric on X

2 Distance on an Ordered Field

Definition 2.1 (Distance). [def:distance] Let F be an ordered field. For $x, y \in F$, the *distance* between x and y is

$$d(x, y) \equiv |x - y|$$

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LaTeX: \dist {x}{y}
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Lemma 2.1 (Symmetry of Distance). [lem:distance-symmetry] *Let F be an ordered field. For all $x, y \in F$,*

$$d(x, y) = d(y, x)$$

Proof. By definition, $d(x, y) = |x - y|$ and $d(y, x) = |y - x| = |-(x - y)|$. It suffices to show $|-(x - y)| = |x - y|$; we split on the sign of $x - y$
Case 1: $x - y \geq 0$. Then $|x - y| = x - y$. Also $-(x - y) \leq 0$, so $|-(x - y)| = -(-(x - y)) = x - y = |x - y|$
Case 2: $x - y < 0$. Then $|x - y| = -(x - y)$. Also $-(x - y) > 0$, so $|-(x - y)| = -(x - y) = |x - y|$ □

Lemma 2.2 (Triangle Inequality for Distance). [lem:triangle-inequality-distance] *Let F be an ordered field. For all $x, y, z \in F$,*

$$d(x, z) \leq d(x, y) + d(y, z)$$

Proof. Let $a = x - y$ and $b = y - z$. Then $x - z = a + b$, so $d(x, z) = |a + b|$. We split on the sign of $a + b$
Case 1: $a + b \geq 0$. Then $|a + b| = a + b$. Since $a \leq |a|$ and $b \leq |b|$,

$$|a + b| = a + b \leq |a| + |b| = d(x, y) + d(y, z)$$

Case 2: $a + b < 0$. Then $|a + b| = -(a + b) = -a - b$. Since $-a \leq |a|$ and $-b \leq |b|$,

$$|a + b| = -a - b \leq |a| + |b| = d(x, y) + d(y, z)$$

□

Corollary 2.1 (Metric on a Complete Ordered Field)**.** [cor:complete-ordered-field-metric] *Let F be a complete ordered field, and let d be the absolute-value distance on F. Under the identification with the reals, (F, d) is a metric space*

Proof. Under this identification, the distance is real-valued. By the definition of absolute value,

$$d(x, y) = |x - y| \geq 0, \qquad d(x, y) = 0 \iff x - y = 0 \iff x = y$$

Symmetry and the triangle inequality give the remaining two metric axioms □

Lemma 2.3 (ε-Criterion for Equality)**.** [lem:epsilon-equality] *Let F be an ordered field. For $a, b \in F$,*

$$a = b \iff \forall \varepsilon \in F_+,\ |a - b| < \varepsilon$$

Proof. In order to proof lem:epsilon-equality, we need to prove both directions
(a): $a = b \implies \forall \varepsilon \in F_+, |a - b| < \varepsilon$
Suppose $a = b$. Then $|a - b| = 0$. For any $\varepsilon > 0$, we have $|a - b| = 0 < \varepsilon$
(b): $\forall \varepsilon \in F_+, |a - b| < \varepsilon \implies a = b$
Assume $a \neq b$
Then, this means that

$$\exists \varepsilon_0 \in F_+, |a - b| = \varepsilon_0$$

Taking $\varepsilon = \varepsilon_0$ gives $\varepsilon_0 = |a - b| < \varepsilon_0$, a contradiction
So, we have $\forall \varepsilon \in F_+, |a - b| < \varepsilon \implies \neg a \neq b$
Since $\neg a \neq b \iff a = b$, So, $\forall \varepsilon \in F_+, |a - b| < \varepsilon \implies a = b$
Therefore, both implications hold, and hence proven □

3 Open Balls

4 Open and Closed Sets

5 Topological Spaces

6 Neighbourhoods

7 Compact Sets

8 Heine-Borel Theorem

9 Cantor Sets